

A-Level Further Mathematics

Cambridge Preparation Notes

Pure Mathematics: Hyperbolic Functions, Matrices,
Differentiation, Integration, Complex Numbers,
Differential Equations

Rayyan Hanif bin Rizal, Kolej MARA Seremban

Contents

1	Hyperbolic Functions	5
1.1	Definitions and Exponential Forms	5
1.2	Fundamental Identities	5
1.3	Differentiation and Integration	6
1.4	Inverse Hyperbolic Functions	6
2	Matrices 2: Eigenvalues, Eigenvectors, and Diagonalisation	9
2.1	Eigenvalues and Eigenvectors	9
2.2	Diagonalisation	10
2.3	Symmetric Matrices and Orthogonal Diagonalisation	11
2.4	Systems of Differential Equations via Matrices	11
3	Differentiation	13
3.1	Implicit Differentiation	13
3.2	Parametric Differentiation	13
3.3	Maclaurin and Taylor Series	14
3.4	L'Hôpital's Rule	15
4	Integration	17
4.1	Integration by Parts	17
4.2	Integration by Substitution	17
4.3	Partial Fractions	18
4.4	Reduction Formulae	18
4.5	Applications of Integration	18
4.5.1	Arc Length	18
4.5.2	Surface of Revolution	19
4.5.3	Volumes of Revolution	19
4.6	Improper Integrals	19
5	Complex Numbers	21
5.1	Modulus-Argument Form and Euler's Formula	21
5.2	De Moivre's Theorem	21
5.2.1	Multiple Angle Formulae	21
5.3	Roots of Complex Numbers	22
5.4	Loci in the Argand Diagram	22
5.5	Complex Exponentials and Trigonometry	23

6	Differential Equations	25
6.1	First-Order ODEs	25
6.1.1	Separable Equations	25
6.1.2	Integrating Factor Method	25
6.1.3	Homogeneous Equations	25
6.1.4	Bernoulli Equations	25
6.2	Second-Order Linear ODEs with Constant Coefficients	26
6.2.1	Complementary Function	26
6.2.2	Particular Integral by Method of Undetermined Coefficients	26
6.3	Variation of Parameters	27
6.4	Euler–Cauchy Equation	27
6.5	Simple Harmonic Motion and Damping	27
6.6	Resonance	27

Chapter 1

Hyperbolic Functions

1.1 Definitions and Exponential Forms

Definition 1.1.1: Hyperbolic Functions

For $x \in \mathbb{R}$:

$$\sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2}, \quad \tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}.$$

The reciprocal functions are $\operatorname{csch} x = 1/\sinh x$, $\operatorname{sech} x = 1/\cosh x$, $\operatorname{coth} x = 1/\tanh x$.

Domain and range: $\sinh : \mathbb{R} \rightarrow \mathbb{R}$ (bijection); $\cosh : \mathbb{R} \rightarrow [1, \infty)$ (even, minimum at $x = 0$); $\tanh : \mathbb{R} \rightarrow (-1, 1)$.

1.2 Fundamental Identities

Osborn's Rule. Hyperbolic identities are obtained from trigonometric identities by replacing $\sin \theta$ with $\sinh x$ and $\cos \theta$ with $\cosh x$, and changing the sign of every product of two $\sinh$ terms.

Key identities (derivable directly from definitions):

$$\cosh^2 x - \sinh^2 x = 1 \tag{1.1}$$

$$\sinh(x \pm y) = \sinh x \cosh y \pm \cosh x \sinh y \tag{1.2}$$

$$\cosh(x \pm y) = \cosh x \cosh y \pm \sinh x \sinh y \tag{1.3}$$

$$\sinh 2x = 2 \sinh x \cosh x \tag{1.4}$$

$$\cosh 2x = \cosh^2 x + \sinh^2 x = 2 \cosh^2 x - 1 = 1 + 2 \sinh^2 x \tag{1.5}$$

$$1 - \tanh^2 x = \operatorname{sech}^2 x, \quad \coth^2 x - 1 = \operatorname{csch}^2 x. \tag{1.6}$$

Derivation of (1.1):

$$\cosh^2 x - \sinh^2 x = \frac{(e^x + e^{-x})^2 - (e^x - e^{-x})^2}{4} = \frac{4}{4} = 1.$$

1.3 Differentiation and Integration

Proposition 1.3.1 Derivatives of Hyperbolic Functions

$$\frac{d}{dx} \sinh x = \cosh x, \quad \frac{d}{dx} \cosh x = \sinh x, \quad \frac{d}{dx} \tanh x = \operatorname{sech}^2 x.$$

Proof: Differentiate the exponential definitions term by term. $\frac{d}{dx} \sinh x = \frac{e^x + e^{-x}}{2} = \cosh x$, and similarly for others.

Standard integrals follow immediately:

$$\int \sinh x \, dx = \cosh x + C, \quad \int \cosh x \, dx = \sinh x + C, \quad \int \operatorname{sech}^2 x \, dx = \tanh x + C.$$

1.4 Inverse Hyperbolic Functions

Theorem 1.4.1 Logarithmic Forms

$$\begin{aligned} \operatorname{arsinh} x &= \ln(x + \sqrt{x^2 + 1}), \quad x \in \mathbb{R}, \\ \operatorname{arcosh} x &= \ln(x + \sqrt{x^2 - 1}), \quad x \geq 1, \\ \operatorname{artanh} x &= \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right), \quad |x| < 1. \end{aligned}$$

Derivation of arsinh: Let $y = \sinh^{-1} x$, so $x = \sinh y = \frac{e^y - e^{-y}}{2}$. Then $e^{2y} - 2xe^y - 1 = 0$. Solving (positive root):

$$e^y = x + \sqrt{x^2 + 1} \implies y = \ln(x + \sqrt{x^2 + 1}).$$

Proposition 1.4.1 Derivatives of Inverse Hyperbolic Functions

$$\begin{aligned} \frac{d}{dx} \operatorname{arsinh} x &= \frac{1}{\sqrt{x^2 + 1}}, \\ \frac{d}{dx} \operatorname{arcosh} x &= \frac{1}{\sqrt{x^2 - 1}} \quad (x > 1), \\ \frac{d}{dx} \operatorname{artanh} x &= \frac{1}{1 - x^2} \quad (|x| < 1). \end{aligned}$$

Derivation: Implicit differentiation. For $y = \operatorname{arsinh} x$: $x = \sinh y$, so $\frac{dx}{dy} = \cosh y = \sqrt{1 + \sinh^2 y} = \sqrt{1 + x^2}$, giving $\frac{dy}{dx} = \frac{1}{\sqrt{1 + x^2}}$.

Standard integral forms:

$$\begin{aligned}\int \frac{dx}{\sqrt{x^2 + a^2}} &= \operatorname{arsinh} \frac{x}{a} + C = \ln(x + \sqrt{x^2 + a^2}) + C, \\ \int \frac{dx}{\sqrt{x^2 - a^2}} &= \operatorname{arcosh} \frac{x}{a} + C \quad (x > a > 0), \\ \int \frac{dx}{a^2 - x^2} &= \frac{1}{a} \operatorname{artanh} \frac{x}{a} + C \quad (|x| < a).\end{aligned}$$

Example 1.4.1: Inverse Hyperbolic Integral

Evaluate $\int_0^2 \frac{dx}{\sqrt{4 + x^2}}$.

Solution. Using $\operatorname{arsinh}(x/2)$:

$$\left[\operatorname{arsinh} \frac{x}{2} \right]_0^2 = \operatorname{arsinh} 1 - \operatorname{arsinh} 0 = \ln(1 + \sqrt{2}).$$

Exercises — Chapter 1

1. Show that $\cosh 2x = 1 + 2 \sinh^2 x$ using the exponential definition.
2. Solve $2 \cosh x + \sinh x = 3$. (Express your answer as a natural logarithm.)
3. Differentiate $y = \operatorname{arsinh}(2x + 1)$ and simplify fully.
4. Evaluate $\int_0^1 \tanh x \, dx$. (*Hint:* $\tanh x = \frac{\sinh x}{\cosh x}$.)
5. **[Extension]** Prove that $\operatorname{artanh}(\tanh 2x) = 2x$ for all $x \in \mathbb{R}$, and use the logarithmic form of artanh to express $\tanh 2x$ as a ratio of exponentials. Verify using the definition.

Chapter 2

Matrices 2: Eigenvalues, Eigenvectors, and Diagonalisation

2.1 Eigenvalues and Eigenvectors

Definition 2.1.1: Eigenvalue and Eigenvector

Let A be an $n \times n$ matrix. A scalar $\lambda \in \mathbb{C}$ is an *eigenvalue* of A if there exists a non-zero vector $\mathbf{v} \in \mathbb{C}^n$ such that

$$A\mathbf{v} = \lambda\mathbf{v}.$$

Any such $\mathbf{v}$ is an *eigenvector* corresponding to λ .

Finding eigenvalues. The equation $A\mathbf{v} = \lambda\mathbf{v}$ has a non-trivial solution iff

$$\det(A - \lambda I) = 0.$$

This is the *characteristic equation*. The polynomial $\chi_A(\lambda) = \det(A - \lambda I)$ is the *characteristic polynomial* of degree n .

Finding eigenvectors. For each eigenvalue λ_i , solve $(A - \lambda_i I)\mathbf{v} = \mathbf{0}$ (row-reduce). The solution space is the *eigenspace* E_{λ_i} .

Example 2.1.1: 3×3 Eigenvalue Problem

Find the eigenvalues and eigenvectors of $A = \begin{pmatrix} 4 & 1 & 0 \\ 2 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix}$.

Characteristic equation:

$$\det(A - \lambda I) = \det \begin{pmatrix} 4 - \lambda & 1 & 0 \\ 2 & 3 - \lambda & 0 \\ 0 & 0 & 5 - \lambda \end{pmatrix} = (5 - \lambda)[(4 - \lambda)(3 - \lambda) - 2] = 0.$$

Inner factor: $(4 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 7\lambda + 10 = (\lambda - 2)(\lambda - 5)$.

Eigenvalues: $\lambda_1 = 2$, $\lambda_2 = 5$ (double).

For $\lambda = 2$: $(A - 2I)\mathbf{v} = \mathbf{0}$ gives $\begin{pmatrix} 2 & 1 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix} \mathbf{v} = \mathbf{0}$, so $v_3 = 0$, $2v_1 + v_2 = 0$.

Eigenvector: $\begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix}$.

For $\lambda = 5$: $(A - 5I)\mathbf{v} = \mathbf{0}$ gives $\begin{pmatrix} -1 & 1 & 0 \\ 2 & -2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, so $v_1 = v_2, v_3$ free. Two independent eigenvectors: $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$.

2.2 Diagonalisation

Theorem 2.2.1 Diagonalisation

An $n \times n$ matrix A is *diagonalisable* iff it has n linearly independent eigenvectors. If P is the matrix whose columns are these eigenvectors, then

$$P^{-1}AP = D = \text{diag}(\lambda_1, \dots, \lambda_n).$$

Equivalently, $A = PDP^{-1}$.

Proof sketch. The k -th column of AP equals $A\mathbf{p}_k = \lambda_k\mathbf{p}_k$, which is the k -th column of PD . Thus $AP = PD$, and invertibility of P (guaranteed by independence) gives $P^{-1}AP = D$.

Matrix powers: $A^n = PD^nP^{-1}$, where $D^n = \text{diag}(\lambda_1^n, \dots, \lambda_n^n)$.

Example 2.2.1: Computing A^n via Diagonalisation

Let $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$. Eigenvalues: $\lambda = 2, 4$ with eigenvectors $\begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

With the modal matrix P and its inverse P^{-1} found as:

$$P = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \quad P^{-1} = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

We can compute A^n by splitting the calculation across lines:

$$\begin{aligned} A^n &= P \begin{pmatrix} 2^n & 0 \\ 0 & 4^n \end{pmatrix} P^{-1} \\ &= \frac{1}{2} \begin{pmatrix} 2^n + 4^n & 4^n - 2^n \\ 4^n - 2^n & 2^n + 4^n \end{pmatrix}. \end{aligned}$$

2.3 Symmetric Matrices and Orthogonal Diagonalisation

Theorem 2.3.1 Spectral Theorem (Real Symmetric Matrices)

If A is a real symmetric matrix ($A = A^T$), then:

- (i) All eigenvalues of A are real.
- (ii) Eigenvectors corresponding to distinct eigenvalues are orthogonal.
- (iii) A is orthogonally diagonalisable: $A = QDQ^T$ where Q is orthogonal ($QQ^T = I$).

Proof (reality of eigenvalues): Let $A\mathbf{v} = \lambda\mathbf{v}$, $\mathbf{v} \neq \mathbf{0}$. Then $\bar{\mathbf{v}}^T A\mathbf{v} = \lambda\|\mathbf{v}\|^2$. Since $A = A^T = \bar{A}$, the left side is also $\overline{\bar{\mathbf{v}}^T A\mathbf{v}} = \bar{\lambda}\|\mathbf{v}\|^2$. Hence $\lambda = \bar{\lambda} \in \mathbb{R}$.

2.4 Systems of Differential Equations via Matrices

A linear system $\dot{\mathbf{x}} = A\mathbf{x}$ (where A is diagonalisable with $A = PDP^{-1}$) is decoupled by the substitution $\mathbf{x} = P\mathbf{y}$, giving $\dot{\mathbf{y}} = D\mathbf{y}$, i.e. $\dot{y}_k = \lambda_k y_k$ for each k . Thus $y_k = C_k e^{\lambda_k t}$ and $\mathbf{x} = P\mathbf{y}$.

Example 2.4.1: Decoupled System

Solve $\dot{x} = 3x + y$, $\dot{y} = x + 3y$ with $x(0) = 1, y(0) = 0$.

From the previous example, $\lambda = 2, 4$, $P = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$. The general solution is:

$$\begin{pmatrix} x \\ y \end{pmatrix} = C_1 e^{2t} \begin{pmatrix} 1 \\ -1 \end{pmatrix} + C_2 e^{4t} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Initial conditions: $C_1 + C_2 = 1$, $-C_1 + C_2 = 0 \implies C_1 = C_2 = \frac{1}{2}$.

$$x = \frac{1}{2}(e^{2t} + e^{4t}), \quad y = \frac{1}{2}(e^{4t} - e^{2t}).$$

Exercises — Chapter 2

1. Find the eigenvalues and eigenvectors of $\begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$.
2. Show that $A = \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix}$ is not diagonalisable over $\mathbb{R}$.
3. If A has eigenvalues λ_1, λ_2 , show $\det A = \lambda_1 \lambda_2$ and $\text{tr } A = \lambda_1 + \lambda_2$.
4. Find A^{10} for $A = \begin{pmatrix} 5 & 3 \\ -2 & 0 \end{pmatrix}$ using diagonalisation.

5. **[Extension]** Let $A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{pmatrix}$. Determine whether A is diagonalisable. If not, explain what obstruction arises. Find A^n for the 3×3 block by writing $A = D + N$ where D is diagonal and N is nilpotent, and use the fact that $DN = ND$ in this case.

Chapter 3

Differentiation

3.1 Implicit Differentiation

If $F(x, y) = 0$ defines y implicitly as a function of x , differentiate with respect to x using the chain rule: wherever y appears, multiply by $\frac{dy}{dx}$.

Example 3.1.1: Implicit Differentiation

Find $\frac{dy}{dx}$ for $x^3 + y^3 = 3xy$ (the Folium of Descartes).

Differentiating: $3x^2 + 3y^2 \frac{dy}{dx} = 3y + 3x \frac{dy}{dx}$.

$$\frac{dy}{dx}(3y^2 - 3x) = 3y - 3x^2 \implies \frac{dy}{dx} = \frac{y - x^2}{y^2 - x}.$$

Second derivatives implicitly. Differentiate $\frac{dy}{dx} = f(x, y)$ with respect to x , again using the chain rule on y terms.

3.2 Parametric Differentiation

For a curve $x = f(t)$, $y = g(t)$:

$$\frac{dy}{dx} = \frac{\dot{g}(t)}{\dot{f}(t)}, \tag{3.1}$$

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\dot{f}(t)}. \tag{3.2}$$

Example 3.2.1: Parametric Second Derivative

For $x = t^2$, $y = t^3$: $\frac{dy}{dx} = \frac{3t^2}{2t} = \frac{3t}{2}$.

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{3t}{2}\right)}{2t} = \frac{3/2}{2t} = \frac{3}{4t}.$$

Arc length (parametric):

$$L = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt.$$

3.3 Maclaurin and Taylor Series

Definition 3.3.1: Taylor and Maclaurin Series

If f is infinitely differentiable at a :

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n.$$

For $a = 0$ this is the *Maclaurin series* of f .

Standard Maclaurin series (valid for indicated domains):

$$\begin{aligned} e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \cdots & (x \in \mathbb{R}), \\ \sin x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} & (x \in \mathbb{R}), \\ \cos x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} & (x \in \mathbb{R}), \\ \ln(1+x) &= \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots & (-1 < x \leq 1), \\ (1+x)^k &= \sum_{n=0}^{\infty} \binom{k}{n} x^n & (|x| < 1), \\ \sinh x &= x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots & (x \in \mathbb{R}), \\ \cosh x &= 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots & (x \in \mathbb{R}). \end{aligned}$$

Combining series. New series are obtained by substitution, differentiation, or integration of known series.

Example 3.3.1: Series by Composition

Find the Maclaurin series for $\ln(\cos x)$ up to the x^4 term.

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \cdots. \text{ Let } u = -\frac{x^2}{2} + \frac{x^4}{24} - \cdots \text{ so } \ln(\cos x) = \ln(1+u) =$$

$$u - \frac{u^2}{2} + \cdots$$

$$= -\frac{x^2}{2} + \frac{x^4}{24} - \frac{1}{2} \left(\frac{x^2}{2} \right)^2 + O(x^6) = -\frac{x^2}{2} - \frac{x^4}{12} + O(x^6).$$

3.4 L'Hôpital's Rule

Theorem 3.4.1 L'Hôpital's Rule

If $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$ (or both $\pm\infty$) and $g'(x) \neq 0$ near a , then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)},$$

provided the right-hand limit exists.

The rule applies repeatedly. Indeterminate forms $0 \cdot \infty$, $\infty - \infty$, 0^0 , ∞^0 , 1^∞ are first converted to $0/0$ or ∞/∞ .

Exercises — Chapter 3

1. Find $\frac{dy}{dx}$ given $e^{xy} + y^2 = x$.
2. For $x = a \cos t$, $y = b \sin t$, find $\frac{d^2y}{dx^2}$ in terms of t .
3. Find the Maclaurin series for $\tan x$ up to and including the x^5 term. (*Hint:* use $\sin x = \cos x \cdot \tan x$ and compare series.)
4. Use L'Hôpital's rule to evaluate $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$.
5. **[Extension]** Derive the Maclaurin series for $\arctan x$ by integrating $\frac{1}{1+x^2}$ term by term. Hence show $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots$ (Leibniz formula), and estimate the error in approximating $\pi/4$ by the partial sum of n terms.

Chapter 4

Integration

4.1 Integration by Parts

$$\int u \, dv = uv - \int v \, du, \text{ or equivalently } \int u v' \, dx = uv - \int u' v \, dx.$$

LIATE priority for u : Logarithms, Inverse trig, Algebraic, Trigonometric, Exponential.

Example 4.1.1: Repeated Integration by Parts

$$I = \int x^2 e^x \, dx. \text{ Apply twice:}$$

$$I = x^2 e^x - 2 \int x e^x \, dx = x^2 e^x - 2(x e^x - e^x) + C = e^x(x^2 - 2x + 2) + C.$$

4.2 Integration by Substitution

Standard substitutions:

Integrand contains	Substitution	Result
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$	Removes square root
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$ or $x = a \sinh t$	
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$ or $x = a \cosh t$	

Example 4.2.1: Trigonometric Substitution

$$\int \frac{dx}{(a^2 - x^2)^{3/2}}, \text{ let } x = a \sin \theta:$$

$$= \int \frac{a \cos \theta \, d\theta}{a^3 \cos^3 \theta} = \frac{1}{a^2} \int \sec^2 \theta \, d\theta = \frac{\tan \theta}{a^2} + C = \frac{x}{a^2 \sqrt{a^2 - x^2}} + C.$$

4.3 Partial Fractions

For a rational function $P(x)/Q(x)$ with $\deg P < \deg Q$, decompose Q into factors:

Factor in $Q(x)$	Partial fraction form
$(x - a)$	$\frac{A}{x - a}$
$(x - a)^n$	$\frac{A_1}{x - a} + \cdots + \frac{A_n}{(x - a)^n}$
$(x^2 + bx + c)$ irreducible	$\frac{Ax + B}{x^2 + bx + c}$

4.4 Reduction Formulae

A *reduction formula* expresses I_n in terms of I_{n-1} (or I_{n-2}), enabling computation for general n .

Example 4.4.1: Derivation: $I_n = \int_0^{\pi/2} \sin^n x \, dx$

Integrate by parts with $u = \sin^{n-1} x$, $dv = \sin x \, dx$:

$$I_n = \left[-\sin^{n-1} x \cos x \right]_0^{\pi/2} + (n-1) \int_0^{\pi/2} \sin^{n-2} x \cos^2 x \, dx.$$

The boundary term vanishes. Using $\cos^2 x = 1 - \sin^2 x$:

$$I_n = (n-1)I_{n-2} - (n-1)I_n \implies I_n = \frac{n-1}{n}I_{n-2}.$$

With $I_0 = \pi/2$ and $I_1 = 1$, this gives Wallis-type results.

Wallis's formula:

$$\frac{\pi}{2} = \prod_{n=1}^{\infty} \frac{(2n)(2n)}{(2n-1)(2n+1)} = \frac{2 \cdot 2}{1 \cdot 3} \cdot \frac{4 \cdot 4}{3 \cdot 5} \cdot \frac{6 \cdot 6}{5 \cdot 7} \cdots$$

4.5 Applications of Integration

4.5.1 Arc Length

For $y = f(x)$ on $[a, b]$:

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx} \right)^2} \, dx.$$

4.5.2 Surface of Revolution

Rotating $y = f(x) \geq 0$ about the x -axis on $[a, b]$:

$$S = 2\pi \int_a^b y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx.$$

4.5.3 Volumes of Revolution

Rotating $y = f(x)$ about the x -axis (disc method):

$$V = \pi \int_a^b y^2 dx.$$

Example 4.5.1: Volume via Disc Method

Volume of sphere radius r : rotate $y = \sqrt{r^2 - x^2}$ about the x -axis on $[-r, r]$:

$$V = \pi \int_{-r}^r (r^2 - x^2) dx = \pi \left[r^2 x - \frac{x^3}{3} \right]_{-r}^r = \frac{4\pi r^3}{3}.$$

4.6 Improper Integrals

An integral is *improper* if the interval is unbounded or the integrand is unbounded on the interval. Evaluate as a limit:

$$\int_a^\infty f(x) dx = \lim_{R \rightarrow \infty} \int_a^R f(x) dx,$$

converges if the limit exists and is finite.

Example 4.6.1: Gamma Function at $n = 1$

$$\int_0^\infty x e^{-x} dx. \text{ Integration by parts: } [-x e^{-x}]_0^\infty + \int_0^\infty e^{-x} dx = 0 + 1 = 1.$$

Exercises — Chapter 4

1. Find $\int x^3 \ln x dx$.
2. Evaluate $\int \frac{3x+1}{(x+1)(x^2+1)} dx$ using partial fractions.
3. Establish the reduction formula $J_n = \int_0^1 x^n e^x dx = e - nJ_{n-1}$, and compute J_2 .
4. Find the arc length of $y = \cosh x$ for $0 \leq x \leq 1$.

5. **[Extension]** Show that $\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$ using the following argument:
let $I = \int_0^\infty e^{-x^2} dx$, write $I^2 = \int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$, convert to polar coordinates, and evaluate.

Chapter 5

Complex Numbers

5.1 Modulus-Argument Form and Euler's Formula

Euler's formula: $e^{i\theta} = \cos \theta + i \sin \theta$.

Every non-zero $z \in \mathbb{C}$ can be written $z = re^{i\theta} = r(\cos \theta + i \sin \theta)$, where $r = |z| > 0$ and $\theta = \arg z$.

Derivation from Maclaurin series: Compare series expansions:

$$e^{i\theta} = 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \cdots = \left(1 - \frac{\theta^2}{2!} + \cdots\right) + i\left(\theta - \frac{\theta^3}{3!} + \cdots\right) = \cos \theta + i \sin \theta.$$

Multiplication and division:

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}, \quad \frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}.$$

5.2 De Moivre's Theorem

Theorem 5.2.1 De Moivre's Theorem

For $n \in \mathbb{Z}$ (and more generally $n \in \mathbb{R}$ for principal values):

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta.$$

Proof (integer n): By Euler's formula, $(e^{i\theta})^n = e^{in\theta}$.

5.2.1 Multiple Angle Formulae

From De Moivre with $z = \cos \theta + i \sin \theta$, let $z^n + z^{-n} = 2 \cos n\theta$ and $z^n - z^{-n} = 2i \sin n\theta$.

Express powers of $\cos \theta$ and $\sin \theta$ in terms of multiples, or vice versa.

Example 5.2.1: Expressing $\cos^4 \theta$

Let $c = \cos \theta + i \sin \theta$. Then $2 \cos \theta = c + c^{-1}$, so

$$(2 \cos \theta)^4 = (c + c^{-1})^4 = c^4 + 4c^2 + 6 + 4c^{-2} + c^{-4} = 2 \cos 4\theta + 8 \cos 2\theta + 6.$$

$$\text{Thus } \cos^4 \theta = \frac{1}{8}(\cos 4\theta + 4 \cos 2\theta + 3).$$

5.3 Roots of Complex Numbers

The equation $z^n = w$ has exactly n roots. Writing $w = Re^{i\phi}$:

$$z_k = R^{1/n} \exp\left(i \frac{\phi + 2k\pi}{n}\right), \quad k = 0, 1, \dots, n-1.$$

These n roots lie equally spaced on a circle of radius $R^{1/n}$ in the Argand diagram.

Example 5.3.1: Cube Roots of Unity

Solve $z^3 = 1$: $z_k = e^{2k\pi i/3}$, $k = 0, 1, 2$.

$$z_0 = 1, \quad z_1 = e^{2\pi i/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i =: \omega, \quad z_2 = e^{4\pi i/3} = -\frac{1}{2} - \frac{\sqrt{3}}{2}i = \omega^2.$$

Key property: $1 + \omega + \omega^2 = 0$, $\omega^3 = 1$.

Sum of roots of unity: For $\omega = e^{2\pi i/n}$,

$$\sum_{k=0}^{n-1} \omega^k = \frac{\omega^n - 1}{\omega - 1} = 0.$$

5.4 Loci in the Argand Diagram

Common loci arising from conditions on z :

- $|z - a| = r$: circle centre a , radius r .
- $|z - a| = |z - b|$: perpendicular bisector of segment ab .
- $\arg(z - a) = \alpha$: ray from a at angle α (excluding a).
- $|z - a|/|z - b| = k$ ($k \neq 1$): circle (Apollonius circle).
- $\arg\left(\frac{z - a}{z - b}\right) = \alpha$: arc of circle through a and b subtending angle α .

Example 5.4.1: Locus Problem

Describe the locus of z satisfying $\left|\frac{z - 2}{z + 2}\right| = 2$.

This is $|z - 2| = 2|z + 2|$, an Apollonius circle. Writing $z = x + iy$:

$$(x - 2)^2 + y^2 = 4[(x + 2)^2 + y^2].$$

Expanding and simplifying: $3x^2 + 3y^2 + 20x + 12 = 0$, i.e. $(x + \frac{10}{3})^2 + y^2 = \frac{28}{9}$.

Circle, centre $(-\frac{10}{3}, 0)$, radius $\frac{2\sqrt{7}}{3}$.

5.5 Complex Exponentials and Trigonometry

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}.$$

These unify hyperbolic and trigonometric functions: $\cos(ix) = \cosh x$, $\sin(ix) = i \sinh x$.

Exercises — Chapter 5

1. Use De Moivre's theorem to express $\sin 5\theta$ in terms of $\sin \theta$.
2. Find all fifth roots of -32 in the form $re^{i\theta}$.
3. Sketch the locus of z such that $\arg(z - 1 - i) = \pi/4$, stating clearly which point is excluded.
4. Show that $\sum_{k=0}^{n-1} \cos(a + kd) = \frac{\sin(nd/2)}{\sin(d/2)} \cos\left(a + \frac{(n-1)d}{2}\right)$ using the real part of a geometric series in e^{id} .
5. **[Extension]** Let $\omega = e^{2\pi i/7}$. Evaluate $\prod_{k=1}^6 (2 - \omega^k)$ by considering the polynomial $z^7 - 1$ factored over $\mathbb{C}$, and substituting $z = 2$. Hence find $\prod_{k=1}^6 |2 - \omega^k|$.

Chapter 6

Differential Equations

6.1 First-Order ODEs

6.1.1 Separable Equations

An ODE of the form $\frac{dy}{dx} = f(x)g(y)$ is separable:

$$\int \frac{dy}{g(y)} = \int f(x) dx + C.$$

6.1.2 Integrating Factor Method

The linear first-order ODE $\frac{dy}{dx} + P(x)y = Q(x)$ is solved by the integrating factor $\mu(x) = e^{\int P(x) dx}$:

$$\frac{d}{dx}(\mu y) = \mu Q \implies \mu y = \int \mu Q dx + C.$$

Derivation: Multiply through by μ : $\mu y' + \mu P y = \mu Q$. The left side is $(\mu y)'$ if $\mu' = \mu P$, i.e. $\mu = e^{\int P dx}$.

Example 6.1.1: Integrating Factor

Solve $\frac{dy}{dx} + \frac{2y}{x} = x^2$.

$P = 2/x$, so $\mu = e^{2 \ln x} = x^2$. Multiply: $(x^2 y)' = x^4$.

$$x^2 y = \frac{x^5}{5} + C \implies y = \frac{x^3}{5} + Cx^{-2}.$$

6.1.3 Homogeneous Equations

$\frac{dy}{dx} = f(y/x)$: substitute $v = y/x$ so $y = vx$, $y' = v + xv'$.

6.1.4 Bernoulli Equations

$\frac{dy}{dx} + P(x)y = Q(x)y^n$: substitute $w = y^{1-n}$ to linearise.

6.2 Second-Order Linear ODEs with Constant Coefficients

The equation $ay'' + by' + cy = f(x)$ has general solution $y = y_c + y_p$, where y_c (complementary function) solves the homogeneous equation and y_p is any particular integral.

6.2.1 Complementary Function

Solve the *auxiliary equation* $a\lambda^2 + b\lambda + c = 0$. Let $\Delta = b^2 - 4ac$.

Case	Roots	Complementary function y_c
$\Delta > 0$	real distinct $\lambda_1 \neq \lambda_2$	$Ae^{\lambda_1 x} + Be^{\lambda_2 x}$
$\Delta = 0$	repeated λ	$(A + Bx)e^{\lambda x}$
$\Delta < 0$	complex $\alpha \pm i\beta$	$e^{\alpha x}(A \cos \beta x + B \sin \beta x)$

6.2.2 Particular Integral by Method of Undetermined Coefficients

$f(x)$	Trial y_p
$P_n(x)$ (polynomial degree n)	$Q_n(x)$ (polynomial degree n)*
e^{kx}	Ce^{kx} *
$e^{kx}P_n(x)$	$e^{kx}Q_n(x)$ *
$\cos kx$ or $\sin kx$	$A \cos kx + B \sin kx$ *

* Multiply by x^s where s is the smallest non-negative integer such that no term of the trial solution is in y_c .

Example 6.2.1: Full Solution of Second-Order ODE

Solve $y'' - 3y' + 2y = e^x$.

Auxiliary equation: $\lambda^2 - 3\lambda + 2 = (\lambda - 1)(\lambda - 2) = 0$. Roots $\lambda = 1, 2$, so $y_c = Ae^x + Be^{2x}$.

Since e^x is in y_c , try $y_p = Cxe^x$:

$$y'_p = Ce^x(1 + x), \quad y''_p = Ce^x(2 + x).$$

Substituting: $Ce^x(2 + x) - 3Ce^x(1 + x) + 2Cxe^x = Ce^x(2 + x - 3 - 3x + 2x) = Ce^x(x - 1 - x) \dots$

Let us redo: $y''_p - 3y'_p + 2y_p = Ce^x(2 + x) - 3Ce^x(1 + x) + 2Cxe^x = Ce^x[(2 + x - 3 - 3x + 2x)] = Ce^x(2 - 3) = -Ce^x$.

So $-Ce^x = e^x \implies C = -1$. General solution: $y = Ae^x + Be^{2x} - xe^x$.

6.3 Variation of Parameters

For $ay'' + by' + cy = f(x)$ with complementary function $y_c = Ay_1 + By_2$, the particular integral is:

$$y_p = -y_1 \int \frac{y_2 f}{W} dx + y_2 \int \frac{y_1 f}{W} dx,$$

where $W = y_1 y_2' - y_2 y_1'$ is the *Wronskian*.

6.4 Euler–Cauchy Equation

The equation $ax^2y'' + bxy' + cy = f(x)$ is solved by the substitution $x = e^t$, transforming it to a constant-coefficient ODE in t .

Key identities under $x = e^t$:

$$xy' = \dot{y}, \quad x^2y'' = \ddot{y} - \dot{y},$$

where dots denote d/dt .

6.5 Simple Harmonic Motion and Damping

The equation $\ddot{x} + \omega^2 x = 0$ (simple harmonic motion) has solution $x = A \cos \omega t + B \sin \omega t$.

The damped oscillator $\ddot{x} + 2k\dot{x} + \omega^2 x = 0$:

- $k < \omega$ (underdamped): $x = e^{-kt}(A \cos \mu t + B \sin \mu t)$, $\mu = \sqrt{\omega^2 - k^2}$.
- $k = \omega$ (critically damped): $x = (A + Bt)e^{-kt}$.
- $k > \omega$ (overdamped): $x = Ae^{\lambda_1 t} + Be^{\lambda_2 t}$, both $\lambda_i < 0$.

Example 6.5.1: Forced Oscillator

Solve $\ddot{x} + 4x = 3 \cos t$, $x(0) = 0$, $\dot{x}(0) = 0$.

$y_c = A \cos 2t + B \sin 2t$. Try $y_p = C \cos t$: $-C \cos t + 4C \cos t = 3 \cos t \implies C = 1$. So $x = A \cos 2t + B \sin 2t + \cos t$. Applying ICs: $A + 1 = 0$, $2B = 0$, giving $A = -1$, $B = 0$.

$$x = \cos t - \cos 2t.$$

6.6 Resonance

If $f(x) = \cos \omega t$ and the natural frequency is also ω (i.e. $\omega^2 = \omega_n^2$ in the auxiliary equation), the trial PI must be multiplied by t , giving *resonance*: amplitude grows without bound as $t \rightarrow \infty$.

Exercises — Chapter 6

1. Solve $\frac{dy}{dx} = \frac{x(1+y^2)}{y(1+x^2)}$ with $y(0) = 1$.
2. Find the general solution of $y'' - 5y' + 6y = e^{2x} + x$.
3. Solve the IVP $y'' + 2y' + 5y = 0$, $y(0) = 1$, $y'(0) = 0$.
4. Use the substitution $v = y/x$ to solve $\frac{dy}{dx} = \frac{y}{x} + \tan\left(\frac{y}{x}\right)$.
5. **[Extension]** Consider $\ddot{x} + \omega^2 x = F \cos \omega t$ (resonant forcing). Assume the general solution takes the form $x = (A + Ct) \cos \omega t + (B + Dt) \sin \omega t$. Substitute into the ODE and show $D = F/(2\omega)$, $C = 0$. With $x(0) = \dot{x}(0) = 0$, write the full solution and describe the long-term behaviour.

Exploration Questions

Complex Numbers and De Moivre's Theorem

- Investigate the geometric shape formed by the roots of $z^n = 1$ in the Argand diagram for different values of n . Describe how symmetry changes as n increases.
- Explore how the expression $(\cos \theta + i \sin \theta)^n$ can be used to generate trigonometric identities for $\cos(n\theta)$ and $\sin(n\theta)$ beyond standard textbook forms.
- Examine what happens to the roots of a complex polynomial when coefficients are slightly perturbed from real to complex values.

Roots of Unity and Loci

- Explore the locus of points satisfying $|z - 1| = |z + 1|$ and extend this idea to $|z - a| = |z - b|$ for complex constants a and b .
- Investigate how the sum of the n th roots of unity behaves when weighted by powers, such as $\sum_{k=0}^{n-1} k\omega^k$.
- Study geometric transformations produced by multiplying a complex number by a fixed root of unity.

Hyperbolic Functions

- Investigate the relationship between hyperbolic identities and circular trigonometric identities using exponential definitions.
- Explore how $\cosh x$ and $\sinh x$ describe the geometry of a catenary curve and compare it with a parabola.
- Study how inverse hyperbolic functions arise naturally in solving integrals involving quadratic expressions.

Differential Equations

- Investigate how different initial conditions affect the family of solutions to $\frac{dy}{dx} = ky$ and interpret this as a growth model.
- Explore the behaviour of solutions to $\frac{d^2y}{dx^2} + \omega^2 y = 0$ under varying ω and connect this to oscillatory motion.
- Study how substitution methods can transform nonlinear differential equations into linear ones.

Polar Coordinates

- Investigate how curves of the form $r = a + b \sin \theta$ change shape as a and b vary.
- Explore conditions under which polar curves intersect themselves and how to detect these intersections analytically.
- Study how converting between Cartesian and polar form changes the complexity of curve equations.

Matrices and Transformations

- Investigate how repeated application of a matrix transformation affects geometric shapes in the plane.
- Explore conditions under which a matrix represents a rotation, reflection, or shear.
- Study eigenvalues as scaling factors and interpret them geometrically in two-dimensional transformations.

Further Investigation

- Explore how different topics in this syllabus connect, such as complex numbers and rotations, or differential equations and exponential functions.
- Investigate whether a single mathematical structure can unify transformations, differential equations, and complex numbers.

“The beauty of mathematics only reveals itself to more patient followers.”

Maryam Mirzakhani